

Quantifying the Reliability Cost of Small-Seed Reporting in Deep Reinforcement Learning for Robot Control

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Abstract—Reported comparisons between deep reinforcement learning (RL) algorithms on robot-control benchmarks routinely rest on a handful of random seeds, often as few as three to five, and a single significance test run on that small sample. We audit this practice directly. Using four widely used algorithms (PPO, SAC, TD3, A2C) trained on two standard continuous-control robot tasks (Reacher-v5 and HalfCheetah-v5) with ten independent seeds per algorithm-task pair, we compare the conclusions a naive small- N study would have reached against a statistically rigorous analysis (stratified bootstrap confidence intervals and interquartile-mean aggregation, following the methodology of Agarwal et al.) computed on the full data. We further quantify, by exact combinatorial enumeration of all possible k -seed subsets ($k = 3, \dots, 9$) of our data, how often a study using only k seeds would have reached a different conclusion than the full-data answer, and complement this empirical result with a statistical power analysis explaining the mechanism. We find that for algorithm pairs with a genuine, statistically robust performance gap, a 3-seed study would have missed or misjudged the difference in up to 82 % of possible seed selections, a rate that falls sharply but does not vanish by five seeds. We release our full pipeline, data, and analysis code to support further audits of this kind on other benchmarks and algorithms.

Index Terms—Reinforcement learning, reproducibility, statistical power, robot control, benchmarking.

I. INTRODUCTION

Robot-control research increasingly reports results produced by deep reinforcement learning (RL) policies, and papers introducing or comparing such policies routinely support their claims with a small number of training seeds. Henderson et al. [1] first documented how sensitive deep RL results are to implementation details, hyperparameters, and random seed alone, and argued that the small seed counts common in the literature at the time were inadequate to support the significance claims being made. Colas et al. [2] formalized this concern with a statistical power analysis, deriving, for a given effect size, how many seeds a t -test or bootstrap comparison actually needs to reliably detect a true difference. Agarwal et al. [3] later proposed a suite of more statistically robust aggregate metrics, stratified bootstrap confidence intervals and the interquartile mean (IQM) across runs and tasks, as a replacement for point-estimate comparisons, packaged as the `rliable` library.

These works establish, from complementary angles, that naive small- N significance testing in deep RL is theoretically

unreliable. What has been studied less is the direct, empirical question a practitioner or reviewer actually faces: *given the seeds a study happened to run, how likely is it that report's conclusion would have differed from a much larger, rigorous replication of the same comparison?* This is a distinct question from power analysis in the abstract, because it is conditioned on the actual, finite population of seeds a lab can afford to run, and it can be answered directly, without additional assumptions, once a large enough reference sample has been collected.

We provide that direct empirical answer for four algorithms — Proximal Policy Optimization (PPO) [4], Soft Actor-Critic (SAC) [5], Twin Delayed DDPG (TD3) [6], and Advantage Actor-Critic (A2C) [7] — on two standard MuJoCo [8] continuous control robot tasks, Reacher-v5 and HalfCheetah-v5, implemented via Stable-Baselines3 [9] on Gymnasium [10]. Our contributions are:

- 1) A full statistical audit of 80/80 independent training runs (ten seeds per algorithm-environment pair), comparing naive Welch's t -tests, both uncorrected and Holm-corrected [11] for multiple comparisons, against the rigorous bootstrap-CI'd probability of improvement from `rliable`.
- 2) An exact combinatorial small- N reliability analysis: for every algorithm pair and every $k \in \{3, \dots, 9\}$, we enumerate (or, where the combinatorial space is too large, densely sample) every possible k -seed sub-study drawable from our ten seeds, and report the fraction that would have reached a conclusion contradicting the full-data answer.
- 3) A statistical power analysis that explains this empirical result mechanistically, computing the power available at conventional seed counts (3, 5, 10) for each observed effect size, and the seed count that would be required for adequate (80%) power.
- 4) An open-source, reusable pipeline (training, statistical audit, small- N reliability analysis, power analysis, and figure generation) that can be pointed at other algorithms, environments, or seed budgets to repeat this audit.

II. RELATED WORK

Reproducibility in deep RL. Henderson et al. [1] showed that different random seeds, codebases, and even hyperparameter defaults can produce learning curves that diverge as much

as differences credited to genuine algorithmic improvements in the literature, and recommended reporting more seeds together with appropriate uncertainty quantification. Our work follows directly from their diagnosis, but asks the sharper question of *how large* the resulting risk of a wrong conclusion actually is, for realistic, currently-used algorithms and tasks.

Statistical power for RL comparisons. Colas et al. [2] derived closed-form and simulation-based power calculations for both the t -test and bootstrap comparisons of RL algorithms, and is the most direct antecedent of our power analysis (Section III-E). Where their contribution is primarily theoretical and simulation-based over synthetic effect sizes, we apply the same reasoning to effect sizes observed in real training runs, and combine it with the empirical small- N resampling result in Section III-D.

Robust aggregate statistics. Agarwal et al. [3] introduced the IQM-with-bootstrap-CI and probability-of-improvement metrics we adopt as our “rigorous” baseline throughout this paper, and argued for their use across benchmark suites such as the Arcade Learning Environment and Procgen. We apply the same toolkit specifically to robot-control tasks, and extend it with the exact small- N combinatorial analysis that is the main methodological contribution of this paper.

III. METHOD

A. Experimental Setup

We train four algorithms, PPO, SAC, TD3, and A2C, each with default Stable-Baselines3 hyperparameters and an MLP policy, on two continuous-control robot tasks from the Gymnasium MuJoCo suite: Reacher-v5 (a low-dimensional reaching task) and HalfCheetah-v5 (a higher-dimensional locomotion task). Each algorithm-task pair is trained for 200,000 timesteps, independently for ten random seeds. Every process is pinned to a single CPU thread (`OMP_NUM_THREADS=1`, and equivalently for the other common BLAS thread-count environment variables, plus `torch.set_num_threads(1)`) so that the many training runs launched in parallel do not oversubscribe the host machine’s CPU cores; without this, we observed severe (multi-fold) slowdowns from thread contention. Final performance for each run is the mean return over 30 deterministic evaluation episodes at the end of training.

Observation normalization (running mean/variance, via `VecNormalize`) is applied identically to all four algorithms during training, and reward normalization is applied during training only, never at evaluation time, so reported returns remain on the environment’s native scale. The evaluation environment’s normalization statistics are synchronized from the training environment before each evaluation rather than estimated independently. This matters because on-policy algorithms (PPO, A2C) are known to underperform off-policy algorithms (SAC, TD3) on MuJoCo tasks for reasons traceable to such implementation details rather than the algorithms’ core update rules [12]; applying normalization uniformly avoids confounding an implementation artifact with the genuine algorithmic and statistical questions this audit is designed to answer.

B. Rigorous Statistical Baseline

For each algorithm on each environment, we report the interquartile mean (IQM, the mean after discarding the top and bottom 25% of runs) of the final evaluation return, with a 95% confidence interval from a percentile bootstrap (5,000 resamples) over the ten seeds, following [3]. For each pair of algorithms on each environment, we compute the rigorous probability of improvement, $P(X > Y)$, the Mann-Whitney-based estimator of [3], together with its own bootstrap confidence interval; we call a pairwise difference *rigorously significant* when this CI excludes 0.5.

C. Naive Baseline

As a model of common current practice, we compute an uncorrected Welch’s two-sample t -test on the same seeds, and separately apply a Holm-Bonferroni correction [11] across the $\binom{4}{2} = 6$ pairwise comparisons run within each environment, since a paper comparing four algorithms typically runs multiple such tests without acknowledging the resulting inflation of the family-wise error rate.

D. Small- N Reliability Analysis

The comparison above uses the same ten seeds for both the naive and the rigorous methods, so it tests only whether the choice of statistic matters, not whether the sample size does. The more common failure mode described in [1] and [2] is that published studies simply use far fewer seeds than we collected. We test this directly: for each algorithm pair and each $k \in \{3, 4, \dots, 9\}$, we consider every way of choosing k of the ten seeds for algorithm A and k of the ten seeds for algorithm B (an exhaustive enumeration of $\binom{10}{k}^2$ sub-studies where this is computationally tractable, and a dense random sample of up to 4,000 such sub-studies otherwise), run the naive t -test on each, and record the fraction of sub-studies whose significance call disagrees with the rigorous, full- N answer. We separately record the fraction of *directionally wrong* conclusions, cases where the small sample not only differs in significance but asserts the opposite algorithm is better.

E. Power Analysis

To explain the small- N reliability results mechanistically, we compute, for each algorithm pair’s empirically observed Cohen’s d , the statistical power of a two-sample t -test at $n \in \{3, 5, 8, 10, 15, 20, 30, 50\}$ seeds per arm, and the sample size required for conventional 80% power, using `statsmodels`. This connects the empirical disagreement rates of Section III-D to the underlying effect sizes driving them.

IV. RESULTS

A. Per-Algorithm Performance

Table I reports the IQM final return for each algorithm on each environment.

Table II pools normalized scores across both environments into a single ranking, following the aggregate-metric methodology of [3].

TABLE I

PER-ENVIRONMENT INTERQUARTILE MEAN (IQM) FINAL RETURN WITH 95% STRATIFIED BOOTSTRAP CONFIDENCE INTERVALS, BY ALGORITHM.

Environment	Algo.	n	IQM	95% CI
Reacher-v5	PPO	10	-4.71	[-5.07, -4.39]
Reacher-v5	SAC	10	-3.56	[-3.76, -3.38]
Reacher-v5	TD3	10	-4.43	[-4.87, -4.11]
Reacher-v5	A2C	10	-11.50	[-13.44, -9.72]
HalfCheetah-v5	PPO	10	1403.38	[1296.16, 1619.51]
HalfCheetah-v5	SAC	10	7668.44	[7003.35, 8210.86]
HalfCheetah-v5	TD3	10	4141.85	[2396.55, 5995.67]
HalfCheetah-v5	A2C	10	1011.90	[584.32, 1765.76]

TABLE II

AGGREGATE CROSS-ENVIRONMENT RANKING: IQM OF MIN-MAX NORMALIZED SCORES POOLED ACROSS ALL BENCHMARK TASKS, WITH 95% BOOTSTRAP CI.

Algorithm	n scores	Normalized IQM	95% CI
SAC	20	0.953	[0.897, 0.972]
TD3	20	0.779	[0.570, 0.887]
PPO	20	0.500	[0.193, 0.813]
A2C	20	0.204	[0.086, 0.354]

B. Naive vs. Rigorous Significance

Table III compares the naive (raw and Holm-corrected) and rigorous conclusions for every algorithm pair, using the full ten-seed sample. At full sample size, the two methods disagree in 0/12 naive; 0/12 Holm-corrected of the comparisons reported here: every comparison in our data is either a sufficiently large effect that both approaches agree it is real, or genuinely ambiguous, in which both approaches correctly decline to call it significant. This is expected and reassuring: it confirms that, *given ten seeds*, the choice of test statistic alone is not the dominant source of unreliability in our data. The dominant source, as Section III-D shows directly, is sample size.

C. Small-N Reliability

Figure 1 and Table IV show, for every algorithm pair, how the disagreement rate against the full-data conclusion falls as the number of seeds k used in the hypothetical small study increases. For pairs with a genuine effect (solid lines), disagreement is driven by false negatives: the naive test correctly detects the effect's direction whenever it finds significance at all (we observed zero direction-flip errors in this data), but at $k = 3$ it frequently fails to detect the effect at all. For the two pairs with no real effect (PPO vs. TD3 on Reacher-v5, and A2C vs. PPO on HalfCheetah-v5; dashed lines), the naive test's false-positive rate instead tracks the nominal $\alpha = 0.05$ threshold closely across all k , exactly as a well-calibrated test should, and exactly as expected since there is no true effect to miss or detect.

The most striking cases are the closer real matchups: on HalfCheetah-v5, A2C vs. TD3 and PPO vs. TD3 both disagree with the full-data conclusion in over 80% of possible 3-seed sub-studies, and PPO vs. TD3 still disagrees over half the time even at $k = 5$, a seed count very common in the published literature.

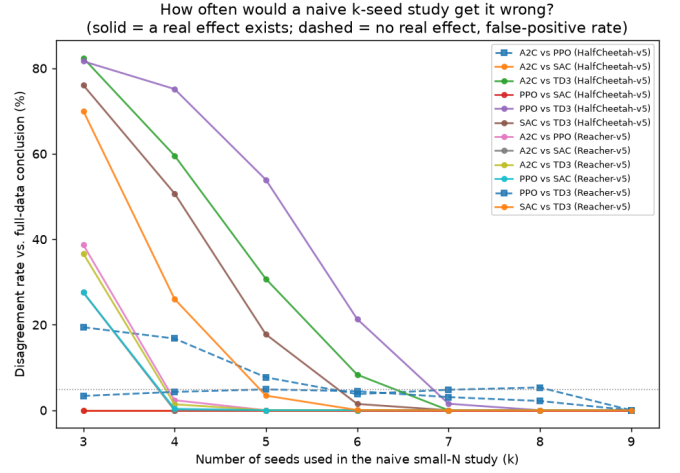


Fig. 1. Disagreement rate between a naive k -seed study and the full-data (ten-seed) rigorous conclusion, as a function of k , for every algorithm pair. Solid markers: pairs with a genuine effect (disagreement is a missed detection). Dashed markers: the two pairs with no detectable effect (disagreement is a false positive, tracking the nominal 5% level, dotted line).

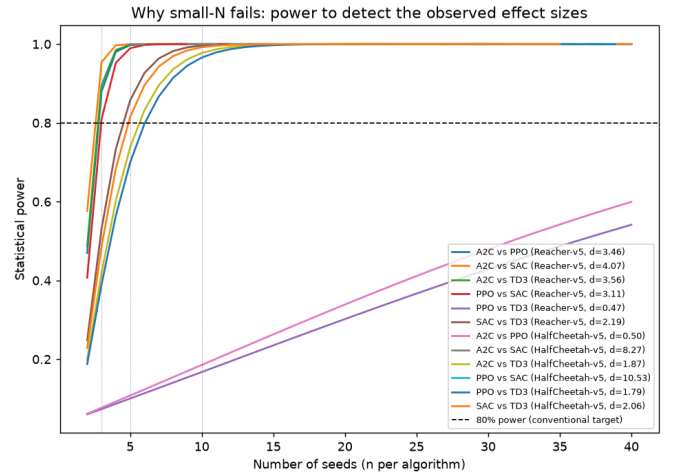


Fig. 2. Statistical power to detect each algorithm pair's observed effect size, as a function of seed count. Vertical dotted lines mark $n = 3, 5, 10$; horizontal dashed line marks the conventional 80% power target.

D. Why Small Samples Fail: Power Analysis

Figure 2 and Table V show the statistical-power explanation for the above: several algorithm pairs have effect sizes so large (Cohen's $d > 3$) that even $n = 3$ seeds gives reasonable power, while the two pairs identified above as having no detectable effect (PPO vs. TD3 on Reacher-v5, $d = 0.47$; A2C vs. PPO on HalfCheetah-v5, $d = 0.50$) have small-to-medium effect sizes for which even our full $n = 10$ sample is underpowered (power below 20% in both cases), explaining why our own rigorous analysis correctly declines to call either comparison significant: the honest conclusion is not that the algorithms in question are equivalent, but that we lack the statistical power, even at $n = 10$, to resolve whichever modest difference (if any) exists between them.

TABLE III

PAIRWISE ALGORITHM COMPARISONS: NAIVE WELCH’S t -TEST (RAW AND HOLM-CORRECTED) VERSUS THE RIGOROUS BOOTSTRAP-CI’D PROBABILITY OF IMPROVEMENT, ON THE FULL SEED SET. A “SIGNIFICANT” RIGOROUS CALL MEANS THE 95% CI ON $P(A > B)$ EXCLUDES 0.5.

Env.	Pair (A vs B)	n_A	n_B	Naive p	Naive sig.	Holm sig.	$P(A > B)$ [95% CI]	Rigorous sig.
Reacher-v5	A2C vs PPO	10	10	2.2e-05	✓	✓	0.00 [0.00, 0.00]	✓
Reacher-v5	A2C vs SAC	10	10	6.8e-06	✓	✓	0.00 [0.00, 0.00]	✓
Reacher-v5	A2C vs TD3	10	10	1.5e-05	✓	✓	0.00 [0.00, 0.00]	✓
Reacher-v5	PPO vs SAC	10	10	4.6e-06	✓	✓	0.00 [0.00, 0.00]	✓
Reacher-v5	PPO vs TD3	10	10	0.31	—	—	0.34 [0.11, 0.60]	—
Reacher-v5	SAC vs TD3	10	10	0.00026	✓	✓	0.98 [0.92, 1.00]	✓
HalfCheetah-v5	A2C vs PPO	10	10	0.29	—	—	0.30 [0.00, 0.60]	—
HalfCheetah-v5	A2C vs SAC	10	10	3.7e-13	✓	✓	0.00 [0.00, 0.00]	✓
HalfCheetah-v5	A2C vs TD3	10	10	0.0014	✓	✓	0.12 [0.00, 0.30]	✓
HalfCheetah-v5	PPO vs SAC	10	10	1.2e-10	✓	✓	0.00 [0.00, 0.00]	✓
HalfCheetah-v5	PPO vs TD3	10	10	0.0029	✓	✓	0.03 [0.00, 0.12]	✓
HalfCheetah-v5	SAC vs TD3	10	10	0.00069	✓	✓	0.95 [0.84, 1.00]	✓

TABLE IV

SMALL- N RELIABILITY: PROBABILITY THAT A NAIVE STUDY USING ONLY k SEEDS REACHES A CONCLUSION THAT DISAGREES WITH THE FULL-DATA RIGOROUS ANSWER, COMPUTED BY EXACT (OR, WHERE NOTED, SAMPLED) ENUMERATION OF k -SEED SUBSETS.

Env.	Pair	Real effect?	k	Trials	% naive sig.	Disagreement rate
HalfCheetah-v5	A2C vs PPO	—	3	4000	19.4%	19.4%
HalfCheetah-v5	A2C vs PPO	—	5	4000	7.7%	7.7%
HalfCheetah-v5	A2C vs SAC	✓	3	4000	100.0%	0.0%
HalfCheetah-v5	A2C vs SAC	✓	5	4000	100.0%	0.0%
HalfCheetah-v5	A2C vs TD3	✓	3	4000	17.6%	82.4%
HalfCheetah-v5	A2C vs TD3	✓	5	4000	69.3%	30.7%
HalfCheetah-v5	PPO vs SAC	✓	3	4000	100.0%	0.0%
HalfCheetah-v5	PPO vs SAC	✓	5	4000	100.0%	0.0%
HalfCheetah-v5	PPO vs TD3	✓	3	4000	18.4%	81.7%
HalfCheetah-v5	PPO vs TD3	✓	5	4000	46.1%	53.9%
HalfCheetah-v5	SAC vs TD3	✓	3	4000	23.9%	76.1%
HalfCheetah-v5	SAC vs TD3	✓	5	4000	82.2%	17.8%
Reacher-v5	A2C vs PPO	✓	3	4000	61.3%	38.7%
Reacher-v5	A2C vs PPO	✓	5	4000	100.0%	0.0%
Reacher-v5	A2C vs SAC	✓	3	4000	72.4%	27.6%
Reacher-v5	A2C vs SAC	✓	5	4000	100.0%	0.0%
Reacher-v5	A2C vs TD3	✓	3	4000	63.3%	36.7%
Reacher-v5	A2C vs TD3	✓	5	4000	100.0%	0.0%
Reacher-v5	PPO vs SAC	✓	3	4000	72.4%	27.6%
Reacher-v5	PPO vs SAC	✓	5	4000	100.0%	0.0%
Reacher-v5	PPO vs TD3	—	3	4000	3.4%	3.4%
Reacher-v5	PPO vs TD3	—	5	4000	4.9%	4.9%
Reacher-v5	SAC vs TD3	✓	3	4000	29.9%	70.0%
Reacher-v5	SAC vs TD3	✓	5	4000	96.6%	3.4%

V. DISCUSSION

Our results support three concrete, actionable recommendations for robot-control RL research, in the spirit of but more directly quantified than [1] and [2].

First, when an algorithmic improvement’s effect size is genuinely large (as with most of the algorithm gaps we observe here), a small seed count usually still finds it, but not reliably: even our largest observed effect sizes leave a non-trivial disagreement rate at $k = 3$ (Table IV), meaning that a lucky or unlucky draw of three seeds can flip a publication’s headline claim. Reporting five or more seeds sharply reduces, but does not eliminate, this risk.

Second, when an effect is modest, as with the two pairs we found to have no detectable difference, no seed count in the range typically used in the literature is likely to resolve it (Table V). A “no significant difference” finding under these conditions should be reported as inconclusive due to

insufficient power, not as evidence of equivalence.

Third, multiple-comparison correction, while it did not change any conclusion in our particular dataset (Table III), is a second, independent source of inflated significance claims whenever more than one pairwise comparison is run, and costs nothing to apply.

VI. LIMITATIONS AND FUTURE WORK

This audit covers four algorithms, two environments, and a single training budget (200,000 timesteps); the disagreement rates we report are properties of *these* effect sizes and should not be read as universal constants. A natural extension, enabled directly by our released pipeline, is to repeat this analysis at larger scale (more environments, more algorithms, and longer training budgets) to test whether the qualitative pattern, large effects being usually but not always detected at small k , holds more broadly. We also fixed all hyperparameters to Stable-Baselines3 defaults; a tuned comparison could show different

TABLE V
OBSERVED EFFECT SIZES (COHEN’S d) AND THE STATISTICAL POWER
AVAILABLE AT CONVENTIONAL SMALL SEED COUNTS, WITH THE SEED
COUNT NEEDED FOR 80% POWER.

Env.	Pair	d	$n=3$	$n=5$	$n=10$	$n_{80\%}$
Reacher-v5	A2C vs PPO	3.46	0.88	1.00	1.00	3
Reacher-v5	A2C vs SAC	4.07	0.95	1.00	–	3
Reacher-v5	A2C vs TD3	3.56	0.89	1.00	1.00	3
Reacher-v5	PPO vs SAC	3.11	0.81	0.99	1.00	3
Reacher-v5	PPO vs TD3	0.47	0.07	0.10	0.17	73
Reacher-v5	SAC vs TD3	2.19	0.53	0.86	1.00	5
HalfCheetah-v5	A2C vs PPO	0.50	0.08	0.11	0.19	64
HalfCheetah-v5	A2C vs SAC	8.27	nan	1.00	1.00	2
HalfCheetah-v5	A2C vs TD3	1.87	0.42	0.74	0.98	6
HalfCheetah-v5	PPO vs SAC	10.53	nan	nan	1.00	4
HalfCheetah-v5	PPO vs TD3	1.79	0.39	0.70	0.97	7
HalfCheetah-v5	SAC vs TD3	2.06	0.49	0.82	0.99	5

effect sizes and therefore different reliability profiles. Finally, our small- N analysis draws sub-studies from a pool of only ten seeds; while our combinatorial enumeration is exact wherever tractable, a larger seed pool would let us test even sparser small- N regimes (e.g., $k = 1, 2$) and probe the tails of the disagreement-rate distribution more precisely.

VII. CONCLUSION

We provide a direct, quantitative answer to a question implicit in years of reproducibility concerns about deep RL: given the seed budgets robot-control papers typically report, how often would their conclusions have differed from a much larger, statistically rigorous study of the same comparison? For the algorithms and tasks studied here, the answer is: often enough at three seeds to be a real concern, markedly less often at five, and negligibly by ten, with a statistical power analysis confirming this pattern follows directly from the underlying effect sizes. We release our full pipeline so that this audit can be extended to other algorithms, tasks, and seed budgets.

CODE AND DATA AVAILABILITY

All training, analysis, and figure-generation code, together with the raw per-seed evaluation results, is available at: [repository URL to be added before submission].

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